Multiplier plus rank-one at $L = 0.3$: a complete reduction, not yet λ_*

PRIME.RDAGGER.LAMBDASTAR_03_KEYSTONE.01 — round 479

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Abstract

At $L = 0.3$ one has $P \equiv 0$, so the compact-window Weil form is a Fourier multiplier plus a rank-one update on each parity: $Q_{\mathrm{W}} = A + \Pi$, $A = (2\pi)^{-1} \int \sigma_A(t)|\hat{h}(t)|^2\, dt$, $\Pi_{\mathrm{even}} = 2\big|\langle h, \cosh(x/2)\rangle\big|^2$. The zero t_c of σ_A and the elementary trace $\mathrm{tr}(P_L B_{t_c} P_L) = 2Lt_c/\pi$ are sealed. The constant minorant $\sigma_A(0)K + \Pi$ is indefinite (interlacing: rank one cannot lift a two-dimensional negative space). Actual A on the even Fourier blocks $N = 2..5$ is positive definite (r478). After five prolates of the band $t_* = 8$ the K-tail is positive with room $c_{\mathrm{tail}} \geq 0.2408$; low-band coupling is $|\sigma_A(0)|\sqrt{\varepsilon} \leq 6.44 \cdot 10^{-3}$. The split closes iff the high-band coupling of $A_+ = \sigma_A\, 1_{|t|\geq 8}$ satisfies $\kappa_{\mathrm{high}} \leq 1.79 \cdot 10^{-2}$. That last operator norm is the exact remainder. Verdict `REDUCED(`$\kappa_{\mathrm{high}} \leq 1.79 \cdot 10^{-2}$ `at` $t_* = 8, N = 5$`)`. This L lies in the Yoshida–Bombieri zone: the value is the method, which must scale to $L = 0.8$. No RH claim.

Status. `REDUCED(`$\kappa_{\mathrm{high}} \leq 1.79 \cdot 10^{-2}$ `at` $t_* = 8, N = 5$`)`. No promotion of an RH claim.

Claim boundary. Nothing here is $\lambda_*(0.3) \geq 0$. The reduction is a theorem; the last coupling is open. $L = 0.3$ is inside the classical YB range $2L < \log 2$: this is a method-and-rigour upgrade, not a new zone. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Convention

$\operatorname{supp} h \subset [-L, L]$, $g = h * \widetilde{h}$, $Q_{\mathrm{W}} = A - P + \Pi$, prime nodes $n \leq e^{2L}$. At $L = 0.3$ the node set is empty, $P \equiv 0$. Plancherel $\hat{h}(t) = \int h(x)e^{itx}\, dx$, $\|h\|_2^2 = (2\pi)^{-1} \int |\hat{h}|^2$. Symbol identity (r476, interval-matched): $\sigma_A(t) = \operatorname{Re}\psi(\frac{1}{4} + it/2) - \log \pi$, even, increasing on $[0, \infty)$. Closed value $\sigma_A(0) = -(\gamma + \log \pi) - (\pi/2 + 3\log 2) \in [-5.372183419225667,\ -5.372183419225665]$.

2 Multiplier plus rank one

Lemma 1 (Even–odd and Π). *A is a Fourier multiplier, hence $A(\mathrm{even}, \mathrm{odd}) = 0$. For real h, $\Pi(h) = \big|\langle h, e^{x/2}\rangle\big|^2 + \big|\langle h, e^{-x/2}\rangle\big|^2$, so on even functions $\Pi = 2\big|\langle h, \cosh(x/2)\rangle\big|^2$ and on odd functions $\Pi = 2\big|\langle h, \sinh(x/2)\rangle\big|^2$. Thus $Q_{\mathrm{W}} = Q_{\mathrm{even}} \oplus Q_{\mathrm{odd}}$. Checked against* `pole_G`$(0, 2L)$*: $\Pi(1) = 16(\cosh L - 1)$.*

Lemma 2 (The zero t_c). *There is a unique $t_c > 0$ with $\sigma_A(t_c) = 0$, and*

$$t_c \in \big[6.289835988836902,\ 6.289835988836904\big]$$

by the digamma enclosure (pad 10^{-16}) plus the r476 identity. Hence $\sigma_A(t) < 0$ on $|t| < t_c$ and $\sigma_A(t) > 0$ on $|t| > t_c$.

3 The time-band operator and the trace ace

Let B_t be the Paley–Wiener projector $(\widehat{Bh})(t) = 1_{|t|<t}\hat{h}(t)$ and $K_t = P_L B_t P_L$ on $L^2[-L, L]$. The kernel is $\sin\big(t(x-y)\big)/\big(\pi(x-y)\big)$, with on-diagonal value t/π.

Lemma 3 (Trace)**.** *$\mathrm{tr}(K_t) = 2Lt/\pi$ exactly. At $t = t_c$, $\mathrm{tr}(K_{t_c}) \in [1.201270186632830, 1.201270186632832]$.*

Nyström (Gauss–Legendre, $n = 16$ vs $n = 48$, top-five difference $< 10^{-14}$, trace match 10^{-8}) gives the Slepian numbers of K_{t_c}

$$\lambda = (0.859391,\ 0.313778,\ 0.027320,\ 7.700 \cdot 10^{-4},\ 1.120 \cdot 10^{-5},\ 1.021 \cdot 10^{-7},\ \dots).$$

Even indices $0, 2, 4$; odd $1, 3, 5$. The first two even eigenvalues already exhaust all but $1.12 \cdot 10^{-5}$ of the even mass. These λ_n are Nyström-stable working constants, not a Lipschitz-interval enclosure.

4 Why the constant minorant fails

Lemma 4 (Minorant, and why it is not enough)**.** *$A \succeq \sigma_A(0)\, K_{t_c}$ as quadratic forms on $L^2[-L, L]$. The rank-one lift $M = \sigma_A(0)K_{t_c} + 2|\cosh\rangle\langle\cosh|$ has $\lambda_{\min}(M) \approx -3.448$ (Nyström). By Cauchy interlacing a rank-one PSD update lifts at most one negative direction. The second even prolate ($\lambda_2 \approx 0.0273$, overlap with* cosh *only* 2.5%*) and the first odd prolate ($\lambda_1 \approx 0.314$, $\|\sinh\|_2^2 = \sinh L - L \approx 0.00452$) remain negative under M. Therefore $M \succeq 0$ is false, and cannot prove $Q_W \succeq 0$.*

The actual form A is much larger than $\sigma_A(0)K$ on every prolate with $\lambda_n \ll 1$ (those functions live outside the negative band). One must use A itself on the essential block.

5 What is already a SATZ

Theorem 1 (Even Fourier blocks, r478)**.** *On* $\mathrm{span}\{\cos(n\pi x/L) : 0 \le n < N\}$ *at $L = 0.3$, the Gram of Q_W is positive definite for $N = 2, 3, 4, 5$: $N = 2$ has $\lambda_{\min} \in [1.731281 \cdot 10^{-2}, 1.825107 \cdot 10^{-2}]$; Higham $\mu_2 \ge 8.302 \cdot 10^{-3}$, $\mu_3 \ge 2.457 \cdot 10^{-3}$. The first odd mode* $\sin(\pi x/L)$ *has $r \ge 2.996973 \cdot 10^{-1}$.*

This is λ_* on a finite-dimensional subspace, not on $L^2[-L, L]$.

6 The reduction

Write $\sigma_A(t) \ge \sigma_A(t_*)\, 1_{|t|\ge t_*} + \sigma_A(0)\, 1_{|t|<t_*}$ for any $t_* > t_c$ with $\sigma_A(t_*) > 0$. Then

$$A \ge \sigma_A(t_*)\, I + \big(\sigma_A(0) - \sigma_A(t_*)\big) K_{t_*}.$$

Let P_N be the projection onto the first N prolates of band t_*, $\varepsilon = \lambda_{N+1}(K_{t_*})$. On $\mathrm{ran}(1 - P_N)$,

$$A \ge \sigma_A(t_*)(1 - \varepsilon) + \sigma_A(0)\varepsilon =: c_{\text{tail}}.$$

Because K_{t_*} is diagonal in this basis, $\langle K_{t_*} P_N h, (1 - P_N)h\rangle = 0$, and the low-band piece of $\langle A\, P_N h, (1 - P_N)h\rangle$ satisfies

$$\kappa_{\text{low}} \le \big|\sigma_A(0)\big|\sqrt{\varepsilon}.$$

The remaining high-band piece is $\kappa_{\text{high}} = \big\|P_N A_+ (1 - P_N)\big\|$ with $A_+ = \sigma_A\, 1_{|t|\ge t_*}$.

Theorem 2 (Exact remainder)**.** *Fix $t_* = 8$ and $N = 5$. Sealed: $\sigma_A(8)_{\text{lo}} \ge 0.240811$, $\varepsilon = \lambda_6(K_8) \le 1.4335 \cdot 10^{-6}$ (Nyström-stable), $c_{\text{tail}} \ge 0.240803$, $\kappa_{\text{low}} \le 6.432 \cdot 10^{-3}$. The even 3×3 Higham floor $\mu_3 \ge 2.457 \cdot 10^{-3}$ gives the numerical budget*

$$\kappa_{\text{high}} \ \le\ \sqrt{\mu_3\, c_{\text{tail}}} - \kappa_{\text{low}} \ \le\ 1.79 \cdot 10^{-2}.$$

If this operator-norm bound is proved, then $\lambda_*(0.3) \geq c$ for an explicit $c > 0$ (Schur of the even 3×3 against a positive tail; the odd sector is strictly easier). The bound is not proved in this round.

The r478 Hilbert–Schmidt majorant of the Fourier off-block is ~ 0.27 and overshoots the budget by an order of magnitude. That majorant is not forced: A_+ on the K-tail is a different operator from the Fourier row-sum.

7 Outlook at $L = 0.8$

There $P \not\equiv 0$ with three nodes $n = 2, 3, 4$. As a quadratic form on h, P is the Fourier multiplier $\sigma_P(t) = \sum 2\Lambda(n)n^{-1/2}\cos(t \log n)$, not a rank-three update of A. Π remains rank two (one per parity). The pair $(A - P, \Pi)$ is still "multiplier plus rank two"; the negative set of $\sigma_A - \sigma_P$ is no longer a single interval, so t_c splits into a union of bands. A four-dimensional Birman–Schwinger picture would be accurate only after replacing P by a finite-rank surrogate. The $L = 0.3$ remainder (κ_{high} of a high-band multiplier) is the piece that must stay explicit if the schema is to scale.

8 Sealed balance

Probe `lambdastar03_probe.py`, SPEC_SHA `7a234db34c2ec6f0`. Smoke 10/10 in 0.36 s; full 22/22 in 0.44 s.

object	status	constant
$\sigma_A(0)$	PROVED	$[-5.372183419225667, -5.372183419225665]$
t_c	PROVED (digamma)	$[6.289835988836902, 6.289835988836904]$
$\mathrm{tr}(K_{t_c})$	PROVED	$[1.201270186632830, 1.201270186632832]$
Slepian $\lambda_n(t_c)$	Nyström-stable	$0.859391, 0.313778, 0.027320, \ldots$
$M = \sigma(0)K + \Pi$	indefinite	$\lambda_{\min} \approx -3.448$
even $G^{(2)}$ $\lambda_{\min}$	PROVED	$[1.731 \cdot 10^{-2}, 1.826 \cdot 10^{-2}]$
even $G^{(3)}$ Higham μ	PROVED	$\geq 2.457 \cdot 10^{-3}$
odd $n = 1$ Rayleigh	PROVED	$\geq 2.997 \cdot 10^{-1}$
$c_{\text{tail}}(8, 5)$	sealed	≥ 0.240803
κ_{low}	PROVED	$\leq 6.432 \cdot 10^{-3}$
κ_{high}	OPEN	need $\leq 1.79 \cdot 10^{-2}$

9 Kill table

gate	status	note
t_c, trace, $\sigma_A(0)$	PASS	every digit a pin
minorant + rank one $\Rightarrow$ PD	KILL	interlacing, $M \prec 0$
even/odd finite blocks	PASS	r478 + odd $n = 1$
tail positivity at $(8, 5)$	PASS	$c_{\text{tail}} > 0.24$
low-band coupling	PASS	$\lvert\sigma(0)\rvert\sqrt{\varepsilon}$
high-band κ_{high}	OPEN	the reduction
row-sum / HS Fourier	KILL	HS $\sim 0.27 > 0.018$
scramble / anti-list	PASS	$P = 0$ vacuous; e^{2L}; no zeros
no RH / not λ_*	PASS	type is REDUCED

10 What this does not do

It does not prove $\lambda_*(0.3) \geq 0$. It does not enclose the Slepian eigenvalues in a Lipschitz interval (Nyström stability only). It does not treat $L > 0.3$. No cofinal quantifier. No RH claim.